

Working file: Data insights and recommendations

Step 3: Analyze and interpret data insights

This template is designed to guide you in analyzing visualized data from SAP Analytics Cloud and formulating actionable recommendations. Use the prompts in each section to support your analysis and ensure a structured approach.

1. Dashboard Overview

Briefly describe the purpose and layout of your dashboard. What KPIs and metrics are displayed?

Purpose: The dashboard is designed to provide SolforemersTech executives with a comprehensive view of operational performance, specifically targeting the decline in sales within the Midwest region, rising production costs at the German factory, and unsatisfactory customer satisfaction scores.

KPIs and Metrics Displayed:

- **Total Sales Revenue:** To monitor overall market performance.
- **Production Cost per Product Line:** To track cost spikes in manufacturing.
- **Customer Satisfaction Score (NPS %):** To measure service quality and customer loyalty.
- **Inventory Turnover Rate:** To identify operational inefficiencies and stock issues.

2. Key Data Insights

List and explain at least three significant insights you have identified from the data visualizations.

- **Insight 1 (Midwest Sales Decline):** According to **Screen Capture 2 (Line Chart)**, there is a consistent downward trend in sales revenue for the Midwest region across the four quarters. While other regions remain stable, the Midwest shows a significant loss in momentum, confirming a loss of market share to competitors.
- **Insight 2 (Production Cost Surge):** **Screen Capture 1 (Horizontal Bar Chart)** reveals that production costs for **Inverters** and **Solar Panels** are disproportionately high in the Southwest and Midwest regions. This indicates that the rising raw material prices from the German factory are severely impacting these high-volume product categories.
- **Insight 3 (Critical Satisfaction Levels):** **Screen Capture 4 (Bullet Chart)** demonstrates that customer satisfaction for Inverters in the Midwest is in the "Low/Red" zone (below 70%). This correlates directly with the sales decline, suggesting that delivery delays and support issues are driving customers away.

3. Trends and Patterns

Describe any noticeable trends, patterns, or anomalies in the data. How do these relate to the business scenario?

Observations: The most critical pattern identified is the strong correlation between declining sales and low customer satisfaction scores, specifically in the Midwest region. The data shows that as satisfaction levels dipped into the "Red" zone for Inverters, sales revenue followed a similar downward trajectory. Additionally, there is a clear upward trend in production costs across high-volume product lines (Solar Panels and Inverters). This anomaly suggests that the operational efficiencies of the German factory are being compromised by external economic factors, leading to a "margin squeeze" where costs rise while sales revenue falls.

4. Strategic Recommendations

Based on your analysis, propose actionable recommendations to address the business challenges.

Recommendation 1: Regional Supply Chain Optimization (Midwest Focus): Establish a local distribution hub or partner with a regional logistics provider in the Midwest to reduce delivery lead times. This directly addresses the "Low" satisfaction scores related to delivery delays.

Recommendation 2: Cost-Effective Sourcing & Manufacturing Review: Conduct a thorough audit of raw material suppliers for the German factory. Consider diversifying suppliers or negotiating long-term fixed-price contracts to mitigate the volatility of energy and material prices.

Recommendation 3: Targeted Customer Support Initiative: Launch a specialized customer success program for the "Inverters" and "Smart Monitors" product lines in underperforming regions. Increasing technical support availability will help recover the Net Promoter Score (NPS) and prevent customer churn to competitors.

5. Expected Impact

Explain the potential impact of each recommendation on the business. Include qualitative or quantitative metrics where possible.

Sales Stability: By addressing delivery delays and service quality, we expect the Midwest sales revenue to stabilize within the next two quarters and potentially grow by 10-15% as customer trust is rebuilt.

Margin Protection: Stabilizing production costs through strategic sourcing will prevent further margin erosion, allowing SolforemersTech to maintain competitive pricing without sacrificing profitability.

Improved Retention: Increasing the NPS from the "Low" (below 70%) to "Medium/High" (above 80%) range will significantly improve customer lifetime value and reduce the costs associated with acquiring new customers.